



INDUSTRIAL CASE STUDY

POWER PLANT CCTV & CORE NETWORK

Design • Deployment • Installation • Commissioning

DOCUMENTED PROJECT SCALE

27

CAMERA / PTZ INSTALLATIONS

2.4 km

FIBRE BACKBONE

US\$6.4K

PROJECT VALUE

Client-confirmed completion evidence: appreciation letter dated 20 February 2026

Prepared as a client-safe technical case study

Vilayat Industrial Estate • Bharuch, Gujarat

SEPTEMBER 2026

From fragmented surveillance to a plant-ready platform

One integrated delivery. Three documented orders. CCTV field deployment, fibre and copper connectivity, rack integration, testing and commissioning were coordinated into one operational outcome.

US\$6,432

PROJECT VALUE

3

DOCUMENTED ORDERS

27

24 FIXED + 3 PTZ

346

FIBRE SPLICES

Business requirement

- Extend surveillance across a live industrial power-plant environment.
- Create resilient optical and structured-cabling pathways.
- Integrate cameras, racks, LIUs, patch panels and field terminations.
- Execute safely, phase-wise, without disrupting plant operations.

Delivered outcome

- End-to-end CCTV installation and commissioning on a fibre-led network.
- Scalable rack and patching architecture.
- Visibility through fixed and PTZ surveillance.
- Client acceptance confirmed in a signed appreciation letter.

Success story

D-Tech translated a mixed civil, passive-network and CCTV requirement into a coordinated industrial deployment.

Field devices were connected through protected HDPE/conduit routes, fibre distribution, Cat6 access links and labelled rack infrastructure. The client acknowledged technical expertise, safety, quality, timely completion and smooth integration.

What Grasim required - what D-Tech delivered

A business-level view of the plant requirement, installed solution and intended result.

CLIENT REQUIREMENT	D-TECH DELIVERY	DELIVERED RESULT
Wider surveillance Fixed and PTZ coverage for designated plant zones.	27 CCTV endpoints 24 fixed and 3 PTZ installations.	Central visibility Live monitoring and controlled evidence review.
Long-distance connectivity Reliable links across distributed industrial areas.	Fibre-led network 2.4 km fibre, 346 splices and protected pathways.	Scalable backbone Plant-ready connectivity for approved growth.
Organized IT core Serviceable racks, patching and structured endpoints.	Structured edge Cat6, LIUs, patch panels, outlets and seven racks.	Simpler maintenance Labelled, traceable infrastructure.
Safe live-plant execution Phased work with minimal disruption.	Testing and handover Commissioning, documentation and snag closure.	Client acceptance Successful integration confirmed in writing.

ONE ACCOUNTABLE DELIVERY PATH

Survey and design → pathways and cabling → racks and cameras → configuration → testing and commissioning → accepted handover

Requirements translated into an industrial architecture

A practical architecture for harsh industrial conditions, distributed endpoints and controlled maintenance.

01 Coverage first

Fixed cameras for defined views; PTZ units for active observation and flexible incident follow-up.

02 Fibre-led backbone

Single-mode fibre distribution reduces distance limitations and supports plant-wide segmentation.

03 Structured edge

Cat6 access cabling, patch panels, I/O jacks, faceplates and labelled outlets simplify change control.

04 Protected pathways

HDPE underground routes and concealed PVC conduit protect cabling across exposed field conditions.

05 Serviceable racks

Floor-standing and wall-mount racks organize LIUs, patching, cable managers and active devices.

06 Verifiable handover

Splice, link, camera and functional checks support traceable commissioning and acceptance.

Design control

Final device locations, fields of view, route lengths, power/network availability and recording policy are validated during site survey and approved installation drawings.

CCTV and IT core integration

Reference architecture based on documented scope; the controlled as-built drawing remains the installation reference.



Optical backbone → copper access layer → central recording and operations

Core areas covered and operating intent

Coverage domains are described functionally to protect client-sensitive location details.

01 Plant perimeter & approaches

Observe external boundaries, approach routes and vulnerable transition zones.

DETERRENCE • DETECTION • REVIEW

02 Controlled access & movement

Support verification of personnel and vehicle movement around controlled entry points.

ACCESS ASSURANCE • TRACEABILITY

03 Power-plant operational zones

Visual oversight of utility and process-adjacent environments without interfering with operations.

SITUATIONAL AWARENESS • SAFETY SUPPORT

04 Transformer / panel endpoints

I/O infrastructure supports connectivity near designated electrical assets.

CONNECTIVITY • FAULT ISOLATION

05 IT / security core & racks

Consolidate fibre, copper, patching and equipment into serviceable distribution points.

AVAILABILITY • MAINTENANCE CONTROL

06 Incident & evidence workspace

Enable authorized teams to view live video, retrieve recordings and investigate.

RESPONSE • EVIDENCE • AUDIT

Privacy and operational controls

Camera views, retention, user privileges and export procedures should follow Grasim's approved security policy and applicable privacy requirements.

Verified installation quantities and materials

Consolidated from Service Orders 4700299647/278 and 4700298966/281 plus Purchase Order 4200334444/281.

INSTALLATION & SERVICES

CCTV

24 fixed + 3 PTZ installations

FIBRE ROUTE

2,400 m fibre cable laying

CIVIL PATHWAY

1,550 m HDPE pipe + 29 m³ excavation

COPPER ROUTE

1,055 m Cat6 + 500 m concealed wiring

FIBRE TERMINATION

346 splices + LIU / fibre patch panels

UTP TERMINATION

110 + 150 terminations / I/O punching

FIELD SUPPORT

5 intermediate poles

RACKS

5 × up to 15U + 1 × 30U + 1 × 15U

KEY PASSIVE MATERIALS

- 2 × loaded 12-port LIUs
- 6 × 24-port Cat6 patch panels
- 200 × 1 m Cat6 patch cords
- 150 × Cat6 I/O jacks
- 100 × dual-port faceplates
- 100 × Cat6 I/O boxes
- 1 × floor-standing 24U rack
- 6 × 1U cable managers
- 2 × LC-LC SM duplex fibre cords
- 15 × PVC boxes with plates

COMMERCIAL REFERENCE

Project value: US\$6,432

USD presentation equivalent consolidated from the three documented project orders.

Technical installation methodology

Phase-wise delivery aligned civil, passive-network, CCTV and commissioning activities in a live industrial environment.

1 Survey & freeze

Validate views, poles, racks, routes, lengths, power, network and permits.

2 Pathway works

Excavate as approved; install HDPE and concealed PVC conduit with protection.

3 Fibre backbone

Pull single-mode fibre, dress into LIUs, splice, protect and identify cores.

4 Rack integration

Install racks, patching, cable managers, uplinks, grounding provisions and labels.

5 Copper access

Lay Cat6, terminate I/O jacks and patch panels, install faceplates and boxes.

6 Camera deployment

Mount fixed/PTZ devices and poles; align views and secure cable entries.

7 Configure & test

Bring links online, validate streams/PTZ, verify time, recording and retrieval.

8 Handover

Close snags, submit tests and as-built references, demonstrate and obtain acceptance.

HSE & QUALITY GATE

Work permits • approved access • barricading • route protection • cable discipline • housekeeping • punch-list closure

Testing, commissioning and acceptance controls

Every layer is tested before operational handover.

TEST DOMAIN	VALIDATION	ACCEPTANCE INTENT
Fibre infrastructure	Continuity, splice quality, core mapping, polarity, LIU dressing and labelled endpoints.	Traceable optical path
Cat6 structured cabling	Wiremap, termination, patching, labelling and outlet-to-rack mapping.	Reliable edge connectivity
Rack & power readiness	Mechanical stability, cable management, segregation, grounding and access.	Serviceable distribution
Fixed cameras	Image, field of view, focus, day/night, stream, timestamp and recording.	Usable video evidence
PTZ cameras	Pan, tilt, zoom, presets, response and operator permissions.	Active observation
Recording & retrieval	Live view, continuity, playback, search, export and authorized access.	Incident readiness
Network integration	Addressing, uplinks, policy alignment where applicable, reachability and error review.	Controlled communications

Acceptance result

The client's appreciation letter confirms successful installation and commissioning, timely completion, smooth plant integration and enhanced security and monitoring capability.

Operational use cases enabled by the solution

The platform supports prevention, response and evidence while providing a scalable network foundation.

Live security monitoring

Centralized visibility for authorized security and operations teams.

Perimeter & access review

Verify movement and investigate exceptions around controlled areas.

Incident reconstruction

Search, replay and export recorded video for investigation.

PTZ situational awareness

Actively follow events from approved operator stations.

Asset protection

Monitor surroundings of designated critical equipment.

Safety support

Review unsafe behavior or events to inform corrective action.

Maintenance diagnosis

Use labelled racks, links and endpoints to reduce fault isolation effort.

Future expansion

Reuse fibre, racks and structured cabling for approved network services.

Governance note

Operational use is subject to authorized roles, retention rules, cybersecurity controls and Grasim-approved procedures.

Why this solution helps Grasim

Benefits reflect the implemented scope and the client's written acceptance.

Wider visibility

Fixed and PTZ cameras extend observation across designated plant zones.

Faster response

Live view and PTZ control help assess events before dispatch.

Evidence quality

Consistent recording and retrieval support incident review.

Lower support effort

Structured racks, labelled patching and mapped endpoints help troubleshooting.

Distance & scale

Fibre distribution supports long plant routes and future growth.

Phased delivery

Coordinated civil, passive and device work reduces disruption.

Safety & quality

Plant-aware execution and punch-list closure support handover.

Expansion ready

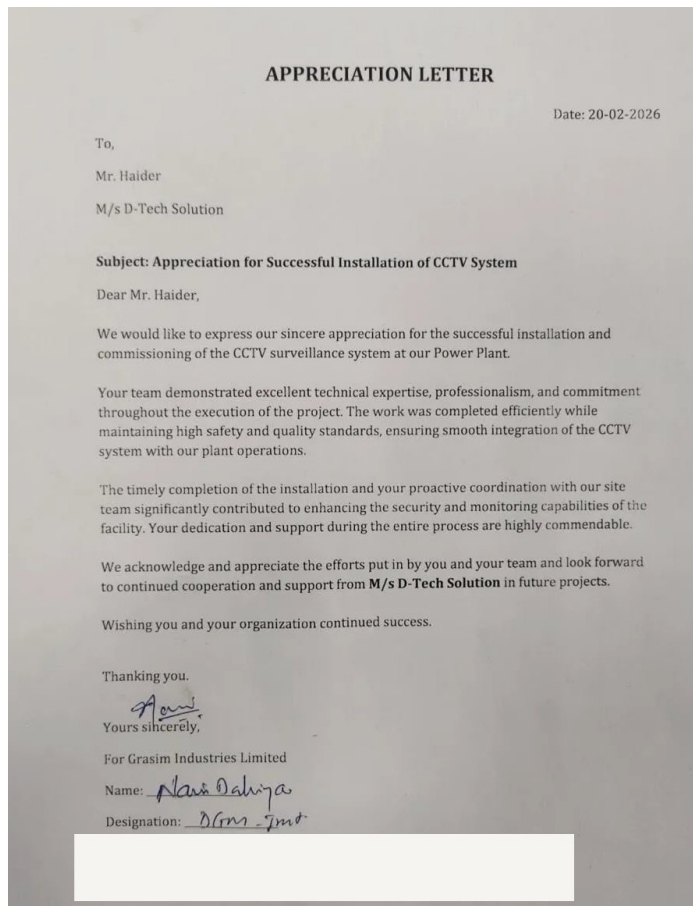
Structured capacity and modular racks reduce rework for additions.

Client voice

The appreciation letter recognizes D-Tech's expertise, professionalism, proactive coordination, safety, quality and timely completion.

Customer appreciation and project references

Signed appreciation letter dated 20 February 2026. Personal contact details are redacted in this presentation copy.



Acceptance highlights

- Successful CCTV installation and commissioning at the Power Plant.
- Technical expertise and professionalism.
- High safety and quality standards.
- Smooth plant integration and timely completion.
- Enhanced security and monitoring.

Document references

SO 4700299647/278 • 21 Feb 2025

SO 4700298966/281 • 08 Feb 2025

PO 4200334444/281 • 08 Feb 2025

Appreciation • 20 Feb 2026

Case study note

Architecture and coverage visuals are conceptual representations. Commercial values are basic values excluding GST.

D-TECH SOLUTION